FACTORY

HIDDEN LAYERS OF AI SYSTEMS

Context, Environment, Organization. Three layers on the road to autonomous engineering.

WHOAMI

```
$ whoami
maker | applied AI engineer @factory.ai
< contextual.ai
< snowflake
< deutsche.bank
# good guy
$ _
```

THE FRAMING

What drives behavior — character, or environment?

Same question for agents.

FROM RESEARCH TO PRACTICE

Different domain. Different layer. Same bet.

DOUWE KIELA

ex-HuggingFace → co-founder, Contextual AI

Co-inventor of RAG. Bet on context engineering for the most complex enterprise knowledge bases.

ENO REYES

ex-HuggingFace → CTO, Factory AI

Bet on the environment and harness for the most complex software engineering. You fell in love with the harness — not the model.

Common conclusion: Systems over models.

WORKING LIST — NOT A CITABLE STANDARD

Seven modes I keep running into. Three I want to dwell on tonight.

01
Hallucination
Fabricated APIs, files, types, names.

02
Context loss
Forgetting earlier turns under window pressure.

03
Task drift
Losing intent over a long-running task.

04
Codebase drift
Many agents, many humans — the codebase stops agreeing with itself.

05
Fake done
Claims complete before validation.

06
Tool misuse
Wrong tool, wrong args, wrong order.

07
Sycophancy
Capitulates instead of pushing back when right.

+
No agreed-upon standard.
Alignment, security, and practitioner taxonomies overlap but don't agree. Be honest about that on stage.

Task drift in a single long-running task.
Codebase drift across a thousand of them.
Same word.
Different time scales.
Different fix.

THE PATTERN STACK THE FIELD CONVERGED ON

Research, plan, implement, verify.

[+ RALPH loop on top]

STEP 01

Research

Discover available design patterns, preferences, AGENTS.md, tool schemas, recent turns. Pull the right context before generation.

STEP 02

Plan

Decompose into ordered actions. Surface the plan before any side effect.

STEP 03

Implement

Act on the environment. Code, run, observe. Cheap, reversible by default.

STEP 04

Verify

External objective signal. Counters the fake-done bias the model has by default.

01 Fake done is a structural bias of any model that wants to please you. Your agent needs back pressure.

02 Session context compression strategy really matters. Every cycle increases the risk of losing intent. At Factory we use anchored iterative summarization.

03 The model that wrote the code is biased toward its own work — it thinks it's the best thing since sliced bread. It is not the most objective validator.

FACTORY SURFACES

Agent readiness
Per-repo assessment & Auto-Fix.

AutoWiki
Continuously refreshed map of the codebase.

Refactoring droids
Architecture ratification at scale.

Security + review droids
External objective signal on every change.

EIGHT PILLARS · CONCRETE SIGNALS WE SCORE

Style & Validation

linter · type checker · formatter · pre-commit hooks · validation under 30s

Without this: 10-min CI loop on a missing semicolon. Agent fixes blind, waits 10 more, repeats.

Build System

single documented command · pinned deps · succeeds on fresh checkout · build < 5 min

Without this: three scripts named build.sh, build_prod.sh, old_build.sh — agent picks wrong one, hits cryptic env-var error.

Testing

unit + integration · runnable locally · unit suite < 2 min · flaky tests fixed or quarantined

Without this: 45-min suite that hits staging DB. Three flakes mask one real failure.

Documentation

AGENTS.md at root · setup/build/test/deploy commands · troubleshooting guide · refreshed in last 90 days

Without this: README points to a wiki updated 18 months ago. Agent finds the working command in a 3-month-old Slack thread.

Dev Environment

devcontainer or CDE · tools/deps/credentials baked in · cold start < 10 min · parity with CI & prod

Without this: Python 3.9 locally, 3.11 in CI. “Works on my machine” — except where it ships.

Code Quality

avg file < 500 lines · functions < 50 lines · cyclomatic < 15 · clear module boundaries

Without this: agent opens a 2,000-line payments.py. Every change has unknown blast radius.

Debugging & Observability

structured logging · distributed tracing · key metrics · errors carry user/request context

Without this: “Payment failed.” That’s the whole log. Agent guesses, redeploys, waits.

Security & Governance

branch protection · secret scanning · CODEOWNERS · dep vuln scanning · CI security gates

Without this: agent inlines an API key, pushes to main. Public repo, security incident.

LONG-HORIZON EXECUTION

Missions

THE ORCHESTRATOR

Plans features, milestones, and the validation contract.

CHILD

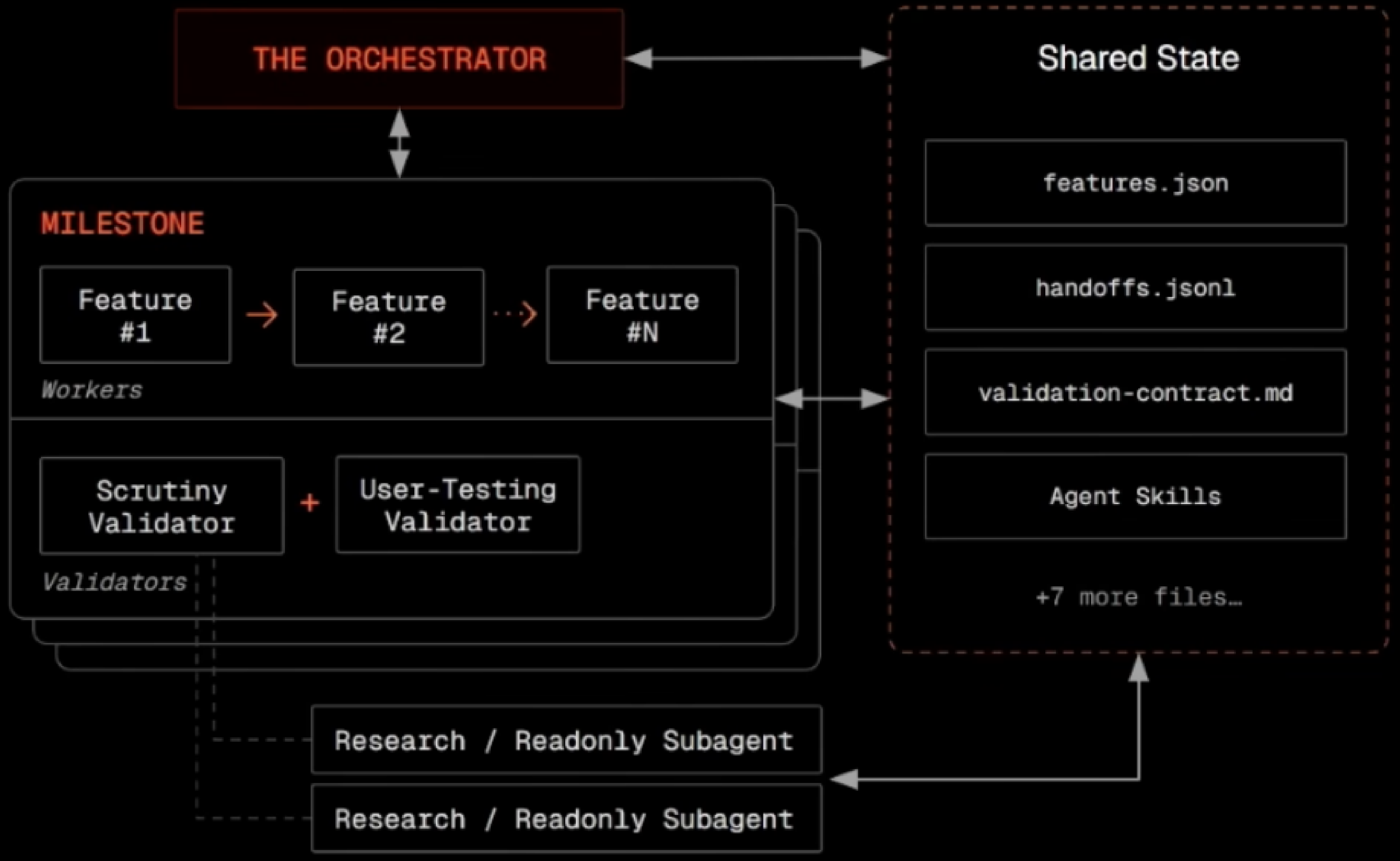
Workers

Fresh context per feature. Implement, commit via git, hand off.

CHILD

Validators

Adversarial verification. Scrutiny + user-testing. Never seen the code before.



Try it today.

```
/missions
```

Describe what you want.

Argue with the orchestrator about scope.

Approve the plan.

Then go do something else.

JOURNEY TO YOUR SOFTWARE FACTORY

You cannot buy a software factory.

01
Constitution
Values, metrics, strategies — non-negotiables every droid in the fleet must honor.

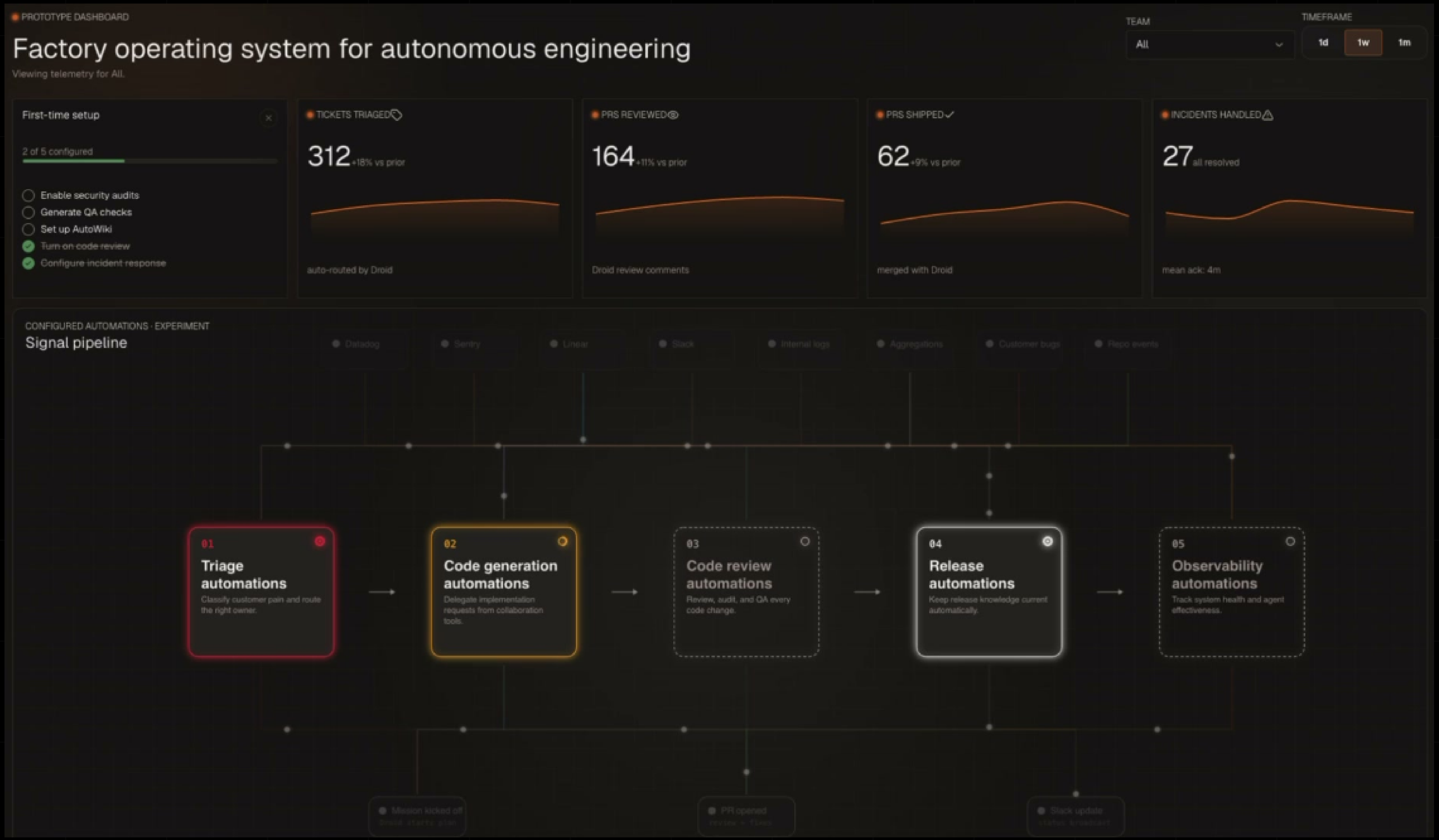
02
Triage rules
Which signal becomes backlog. Which item becomes a mission.

03
Red lines
What no droid ships without a human. Encoded, not folklore.

04
Continuous learning
Decision history, frictions, taste — flowing back into the constitution.

05
Signal → problem → triage → resolution → validation → deployment → reflection
The full loop, encoded as missions and contracts — not folklore.

You build it. Invest in your organizational knowledge. Be the agent of change.



—— THREE LAYERS. THREE FAILURES. THREE TREATMENTS.

From writing code to designing the machine that builds the machine.

Models keep getting smarter. The leverage — and the moat — stay in the layer and the knowledge around them.

Thank you — let's talk.

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